Westlivingster – Executive Summary of the Business Questions

Data Pack Resource A.3 · From the Board of Directors of Westlivingster Ltd · Coursework Part A, 5COSC020W Database Systems 2026/27

Dear Data Architects,

Thank you for the many Business Questions you posted on our forum between 5 and 15 October 2026. Many of you asked similar questions, so rather than replying to each post, we have grouped our answers by theme below. These answers complete our Project Brief (Data Pack Document A.1): where they add detail to the brief, please follow them. Each answer is numbered, so you can refer to it in your work (for example, “BQ answer 12”).

We look forward to seeing your conceptual EERDs.

The Board of Directors, Westlivingster Ltd

A. Residents, tenants and owners

1. Anyone who registers with us is one of our residents from that moment, whether or not they go on to rent or buy, and their details stay on our records. When registering, each person chooses either to rent or to buy. A resident is either a tenant or an owner, never both.
2. If one of our tenants later decides to buy an apartment, they register with us again, this time as someone who wishes to buy. Their registration as a tenant, and all their past rental agreements, are kept as they were. In the same way, anyone who changes their mind between renting and buying registers again.
3. Our owners live in the apartment they own, so an owner owns one apartment only, and does not let it out to anyone else. Only the person who purchased the apartment is registered as its owner, even if they bought it with someone else.
4. We only need to know who owns each apartment now, and the date they bought it. When an owner sells their apartment, the new owner registers with us and replaces the previous one.
5. Only the people who register with us are recorded: family members or others living with them are not, unless they register themselves.
6. For residents and tenants, we only keep the current details: if someone’s contact details change, or a tenant’s salary or credit score is updated, we simply update the record.

B. Rental agreements

7. Each rental agreement is for one apartment only. A tenant has only one current rental agreement at a time, as they live in the apartment they rent. Over time, however, a tenant can appear on many agreements, for example after renewals, or when they move to another of our apartments or come back to us after some time away, and we keep all of them. Their registration with us stays the same throughout. In the same way, an apartment has at most one current rental agreement, but it will have many agreements over the years, and we keep them all.
8. Every agreement has exactly one lead tenant, who is always one of the tenants who sign it. When a tenant signs alone, they are the lead tenant. When several tenants sign together, they choose among themselves who will be the lead tenant, and we record their choice.

9. The tenants on an agreement, its lead tenant and its start and end dates stay the same for the whole agreement, and apply to all its tenants. If a tenant leaves or joins, or the lead tenant changes, the current agreement ends and a new one is signed. The same applies when a tenancy is renewed: the renewal is a new agreement, with its own dates, deposit and total cost. In every case we keep the previous agreement with all our past agreements.
10. If an agreement ends early, we record the date it actually ended, and its utilities contracts end with it.

C. Utilities

11. Westlivingster chooses the suppliers and sets up the utilities contracts for each rental agreement when it is signed, so every agreement has at least one utilities contract, usually several. Each new agreement, including a renewal, has its own new contracts. Each contract is with one supplier, who may supply one or several utilities under it, and covers one rental agreement only, for all the tenants on that agreement. A supplier can of course have contracts for many different agreements.
12. For each contract we record its supplier, its start and end dates, and its estimated cost for the whole contract period. This estimate is recorded separately and is not part of the total cost of the tenancy. If the supplier changes during a tenancy, the old contract ends and a new one starts, and we keep both.
13. Owners arrange their own utilities, so we do not record them.

D. Apartments, buildings, facilities and parking lots

14. An apartment can be empty, with no current tenants and no owner. For each apartment we record its reference number, its number of bedrooms, a short description, including its size, and its monthly rent. We only keep its current monthly rent: if it changes, we simply update it.
15. Each building contains at least one apartment and offers at least one common facility; it may offer several, including more than one of the same kind. Each facility belongs to one building only. For each facility we record its reference number, its name and a short description. We do not record who uses the facilities, bookings, opening hours or fees.
16. Each parking lot belongs to one building. A building may have one or several lots, or none at all; residents of a building without its own parking may be given a space in another building's lot. Each lot is either for residents' apartments or for visitors, never both, and is divided into a number of spaces. There is no other parking at Westlivingster.

E. Parking for residents

17. Each apartment can have at most one parking space, however many people live in it. The space is given to the apartment, not to a person, and is a specific, numbered space reserved for that apartment. It is usually in a lot belonging to the apartment's building, but as not every building has its own parking, it may be in another building's lot. Owners can request a space with their apartment in the same way as tenants.
18. Residents who have been given a space tell us which one of their vehicles will use it. Only one vehicle is registered to the space at any time; if they change it, for example after buying a new car, they simply tell us and we record the new one. If their new vehicle is fully electric, they must ask for a space with a charging point.

19. We only need to know the current situation: which apartment has which space, and which vehicle uses it. We do not keep a history of past allocations or past vehicles, and we do not record requests that were refused or are still waiting. If residents move to another apartment, the space stays with the apartment they leave; they can request a space with their new apartment.
20. Residents' spaces are for residents' vehicles only; visitors use visitor spaces.

F. Vehicles, electric vehicles and hybrids

21. Each vehicle is registered with us to one person only: either one of our residents or one visitor, never both. A car shared by a couple or a family is registered to just one of them. Residents and visitors may each have several vehicles registered with us; there is no limit. If a vehicle belongs to a company, we register it to the person who uses it.
22. Every vehicle runs on electricity, on petrol or diesel, or on both. A hybrid runs on both, so for a hybrid we record both its type of electric vehicle and the fuel it uses; there is no separate category for hybrids, and we treat all hybrids the same way, whether or not they plug in.
23. Residents with a fully electric vehicle must request a space with a charging point. A resident with a hybrid may request either a space with a charging point or a standard space, but uses only one space. Visitor spaces do not have charging points.

G. Visits

24. A visit is one visitor coming to one apartment. If a visitor goes to two apartments, we record two visits; if several visitors come together, we record a visit for each of them. We record which apartment was visited, not which resident.
25. We record a visitor's details once, the first time they come. When they come back, whether to the same apartment or a different one, we simply record the new visit. Regular and occasional visitors are treated the same way. Each time a visitor arrives, we record a new visit with the date and time it started and ended, and we keep every visit.
26. We only record visits from people who do not live at Westlivingster. When residents visit each other, we do not record it.

H. Parking for visitors

27. Visitors register their own vehicle for one of our visitor spaces when they arrive. Each use is of a specific, numbered space, but visitor spaces are not reserved for particular visitors or apartments: a visitor may use a different space each time, and the space may be in any of our visitors' lots.
28. A visitor's vehicle is registered with us once; if they come back with the same vehicle, we reuse its details, and they may also come with different vehicles on different visits. We record every use of a visitor space: which vehicle used which space, and when it arrived and left. If a vehicle moves to another space, that is recorded as a separate use.
29. We record visits and the use of visitor spaces separately: not all our visitors come with a vehicle. We do not need to link a parking record to a particular visit.

I. Reference numbers and details we record

30. We give every resident, visitor, rental agreement and utilities contract its own reference number. Vehicles are identified by their registration number, and we also record their make, model and a short description. For residents and visitors, we record their full name and contact details (email address and mobile number); for residents, we also record their date of birth. For each utilities contract, we only need the supplier's name.

J. Matters outside the scope of this project

31. Several questions concerned matters such as rent payments and arrears, deposits, fees and charges, notice periods, maintenance, staff, waiting lists and what happens when no parking space is available, time limits for visitors, legal matters, how long we keep personal data, and business reports and statistics. These are handled outside the system you are designing, so you do not need to record them. The same applies to rare situations not covered above, such as a visitor who later becomes one of our residents.

32. Some questions asked how to model our data, for example which entities, specialisations or keys to use. As a Board, we are not database specialists: please raise questions about modelling techniques with your tutors in your tutorial sessions.

A reminder from the module team

As Question 2 states, your conceptual EERD must show the entities with their attributes and primary keys, the specialisations with their constraints, the relationships with their multiplicities on both ends, and any attributes on relationships. Nothing else belongs in a conceptual model: in particular, foreign keys must not appear.